ISPRL: MANAGING INDIA'S SOVEREIGN CRUDE OIL PRICE RISK

Learning Business Outside the Classroom

On the first day of the second semester of the MBA program at a Business School in 2022, a young professor gave the class the option of taking 'Learning Business Outside Classroom' (LBOC), a non-classroom credit course, instead of a conventional subject. Students were expected to identify a business or institution, understand its purpose and value proposition, interact with senior management and submit a report and presentation. Nearly 20 students expressed interest; one of them was Basavraj, who welcomed the opportunity to study an organization beyond the classroom. The young professor encouraged them to select unusual organizations so that the entire class could learn from each project.

Basavraj, from Mangalore, Karnataka, had long been fascinated by Indian Strategic Petroleum Reserves Limited (ISPRL) and its nearby underground crude-oil facility. Photographs of enormous rock caverns on the company's website convinced him that the project combined engineering, finance, energy security and public policy. He formed a team of three other students and began researching strategic petroleum reserves (SPRs) around the world, their construction and maintenance costs, and their emergency role. Then, he wrote to H. P. S. Ahuja, the ex -Managing Director of ISPRL, and was invited to interact with the management team at Mangalore.

India's crude oil import dependence reached a reported 88.7% in FY 2025–26. The project that Basavraj's team selected remains an important way to examine the economic cost of dependence on imported oil. See Exhibits VI and VIII.

Strategic Emergency Reserves and the International Energy Agency

The International Energy Agency (IEA) was established in 1974 in response to the 1973–74 oil shock. IEA was meant to support oil-supply security through emergency preparedness, common information systems, analysis, energy efficiency, sustainability and coordinated response to serious disruptions. As of October 2026, the IEA had 32 member countries, 14 Association countries and six countries in the accession process. India joined as an Association country in 2017 and is not yet a full member.

An essential requirement for gaining IEA membership was that the emergency stocks of crude oil and/or refined petroleum products should be equivalent to at least 90 days of the previous year's average daily net oil imports, accessible to the government during supply disruptions. Member countries were also mandated to maintain

Instructions for Students /Participating Teams
*The case and its accompanying exhibits are provided exclusively for the **AIMA–ICRC India Case League 2026**. Students and participating teams shall use the information solely for preparing their competition presentations, developing recommendations, and participating in case discussions and evaluations.*

***Strict adherence to the data and information provided in the case and its exhibits is mandatory.** Teams must base their analyses, arguments and recommendations exclusively on the information supplied in the case materials. **No external data, updated figures, subsequent developments, additional information or modifications to the provided data will be accepted** during case presentations or discussions.*

measures to restrain demand by up to 10%. Notably, while the 90-day stock requirement applied to IEA members, it did not automatically bind India in its association-country capacity. Nor is it the same as 90 days of domestic consumption.

Emergency stock may be owned by a government, a designated stockholding agency or the oil industry; countries combined these arrangements to meet their requirements. An agency could hold inventory directly, whereas industry-held stocks were maintained by refiners and other operators subject to national obligations. Countries also had the option to arrange for stocks to be held abroad under enforceable agreements that ensured physical access when needed. Whatever the ownership structure, emergency release required authorization from the relevant public authority. Such stocks were meant to protect against severe physical supply interruption and the economic damage from disruption; they were not simply a trading portfolio for profiting from price movements (Exhibit I).

The IEA could organize collective action when a large disruption threatened the global market. In the past, collective responses in 1991, 2005 and 2011 were reported. By October 2026, there had been six such actions: those three, two in 2022, and the decision on 11 March 2026 to make 400 million barrels available after Middle East supply disruptions. This was the largest announced collective action in IEA history (Exhibit II).

Costs Associated with Construction and Maintenance of Strategic Reserves

Strategic reserves required substantial investment in storage caverns or tanks, pipelines, marine facilities, land, safety systems and oil inventory. The three broad physical storage included underground rock caverns, underground salt caverns and above-ground steel tanks. Each had different geological, technical, environmental and security implications; for instance, rock caverns required appropriate rock formations and groundwater conditions; salt caverns required suitable underground salt geology and management of saline brine; above-ground tanks demanded more surface area and could be exposed to different hazards.

The historical construction-cost ranges of US$15–31 per barrel for rock caverns, US$8–12 for salt caverns in favorable brine-disposal conditions, and US$29–33 for above-ground storage; and the reported illustrative historic operating and maintenance costs of less than US$0.40 per barrel for underground caverns compared with as much as US$3 per barrel for some newly built above-ground facilities. These were historical planning estimates rather than quotations available in 2026. Actual bids depended on construction materials, geology, regulations, interest rates, currency and supporting infrastructure (Supplementary Exhibit S1).

Buying crude oil to fill a cavern would be more financially demanding than constructing the cavern itself. Unlike an ordinary warehouse, a strategic store tied up inventory capital that could otherwise fund competing public priorities. Refined products could also need periodic rotation to maintain specifications. Ownership, inventory financing and guaranteed access therefore mattered as much as installed cubic meters of rock cavern.

> *The Government's 2021 approval of 6.5 million metric tons (MMT) of additional storage remains a project commitment, not an addition to commissioned capacity. The latest published ISPRL FY 2024–25 annual report records major Padur Phase II procurement and land milestones and a continuing land-allotment process for Chandikhol; these must not be counted as completed facilities.*

India's Strategic Petroleum Reserve Program

India began considering strategic oil reserves in the late 1990s, and incorporated Indian Strategic Petroleum Reserves Limited (ISPRL) in June 2004. Since May 2006, ISPRL has been a wholly owned subsidiary of the Oil Industry Development Board (OIDB); it functioned under the Ministry of Petroleum and Natural Gas (MoPNG). ISPRL's role was to develop and operate India's dedicated underground strategic crude-oil storage facilities.

Under Phase I, three underground rock-cavern facilities were completed at Visakhapatnam in Andhra Pradesh, and Mangalore (Mangaluru) and Padur in Karnataka. The combined installed capacity was 5.33 MMT:

Visakhapatnam 1.33 MMT, Mangalore 1.50 MMT and Padur 2.50 MMT. The commissioning program extended from 2015 to 2018; the installations were dedicated to the nation in February 2019. The sites were selected with reference to geology, ports and refining/pipeline access (Exhibits III–V).

To augment capacity, the government approved an additional 6.5 MMT under Phase II in July 2021: 4.0 MMT at Chandikhol in Odisha and 2.5 MMT at Padur in Karnataka, intended as commercial-cum-strategic facilities under a public-private partnership (PPP) framework. The ISPRL annual report for FY 2024–25 recorded that 204 acres at Padur had been handed over in July 2025; the request for proposals had been issued in December 2024; a letter of award was issued on 18 August 2025; and the target completion date was August 2030. For Chandikhol, Odisha had given an in-principle approval in January 2025 and signed a memorandum of understanding (MoU) in April 2025; while the land-allotment process remained ongoing. Phase II was therefore not part of India's operating reserve at the 2026 case date.

The distinction between capacity and oil actually held became especially important in 2026. An official government statement in March 2026 put total Indian petroleum-storage capacity at approximately 74 days and actual stock cover at approximately 60 days, including crude oil, petroleum products and dedicated cavern inventory. These figures covered India's broader oil system and were not statements that ISPRL's own caverns held 60 days of crude. The financial-year 2024–25 report separately disclosed about 3.52 MMT of crude in the ISPRL caverns as of 31st March 2025; such figures could not be interchanged (Exhibit VI).

Government Strategy for Filling ISPRL's Phase I Facilities

The Government of India adopted two broad approaches: direct government financing of crude inventory, and storage arrangements with producing-country companies. During the exceptional price collapse in April–May 2020, the government purchased approximately 16.71 million barrels of crude at an average price close to US$20 a barrel, using the opportunity to stock its caverns more economically. These purchases remained as historical evidence of the potential benefits of opportunistic filling; they were not a fixed model for future purchases at uncertain prices.

Under a 2018 arrangement, Abu Dhabi National Oil Company (ADNOC) was permitted to store crude in a cavern at Mangalore. The FY 2024–25 ISPRL annual report described an initial fill of roughly 5.8 million barrels and the provision that part of the oil could be used commercially while the balance remained subject to emergency access rights. The report also recorded a September 2024 MoU to discuss extension of cooperation and potential interest in storage at Visakhapatnam. Commercial rights and sovereign rights therefore coexisted in the model; a storage arrangement could save upfront government inventory expenditure but also creates contractual dependencies.

On 15th May 2026, there was a further development occurred: ISPRL and ADNOC exchanged a strategic collaboration agreement. Official statements described opportunities for potential ADNOC storage of up to 30 million barrels in India, across sites in Visakhapatnam and Chandikhol, and possible crude storage in Fujairah, UAE, as part of India's strategic petroleum reserve. These were potential areas for collaboration, not evidence that the indicated volumes had already been contracted and filled (Supplementary Exhibit S2).

Basavraj's Team Discussion with ISPRL Officials at Mangalore

The following questions and answers were meant to preserve the substance of the historical management interaction underlying the original 2022 interview. They should be read as interview evidence from that period in order to help understand 2026 issues.

Why does India need strategic reserves?

The officials explained that reserves formed part of India's energy-security policy, particularly because domestic crude production met only a small part of national requirements. Severe import-supply disruption, unexpected

political instability, conflict or sharp price increases could harm the economy. The purpose of a reserve was to give the country time to respond, rather than eliminate exposure to international petroleum markets.

FY 2025–26 crude import dependence of 88.7% reinforces the underlying exposure; it replaces the older generic formulation that local production met less than 15% of requirements.

Why were the locations selected?

The officials described three considerations: proximity to a seaport for receiving imported oil, geological conditions suitable for underground rock caverns, and access to refineries through pipelines. Visakhapatnam supported supply access on the east coast; Mangalore and Padur supported the southern and western petroleum logistics network. Future locations, including Chandikhol and the Padur extension, were assessed on comparable physical and logistical criteria.

Why underground rock caverns rather than salt caverns or steel tanks?

According to the officials, underground rock caverns offered security, longevity, less exposure at the surface, and suitable operating economics for India's geology. Salt caverns could be inexpensive to excavate under the right conditions but posed brine-management challenges. Steel tanks on the other hand, required more land and a different safety and maintenance arrangement.

How do the caverns operate?

The officials described the scale of construction, tunnels, individual cavern compartments, pipelines and marine unloading arrangements. Crude delivered by ship would be moved through dedicated pipelines to the caverns after appropriate safety preparations. Different crude grades could be stored separately. Emergency withdrawal would require pumping and transfer to refineries or other suitable points through connected infrastructure. The ability to move crude reliably was an essential part of the reserve's value, not merely the quantity of oil stored.

Who bears construction and inventory costs?

The discussion distinguished capital expenditure on the underground facility from funding the crude held within it. Phase I construction was publicly funded. For subsequent development, government agencies considered the PPP model in which private parties could develop facilities in exchange for contractual operating rights. The older 3 facilities used design-build-finance-operate-transfer (DBFOT) concepts; the later Padur Phase II procurement milestone made PPP implementation more concrete, although the completed capacity had not yet been delivered by the case date.

Why would ADNOC store oil in India?

The officials observed that India was an important customer market for oil producers. Storage near Indian refineries could support sales and logistical convenience while reducing the government's need to fund every barrel of reserve inventory directly. At the same time, any arrangement relied on the specific commercial rights, emergency access provisions, replenishment obligations and international relationship. The 2026 collaboration announcement extended that question to storage opportunities both inside and outside India.

How are safety, insurance and environmental risks managed?

The officials described operating standards, safety checks, firefighting systems, hydrocarbon and smoke detection, environmental safeguards and insurance coverage. The case recorded reliance on Oil Industry Safety Directorate (OISD) guidance and quality, health, safety and environmental (QHSE) systems. These arrangements remained important because a storage facility unable to operate or draw down safely during an emergency would have limited strategic value.

Who determines the release and price of reserve oil?

The interview noted that emergency release decisions rested with the Indian government. The officials discussed possible releases to oil refineries and the government's role in authorizing strategic stock use. The wider international experience showed that countries differed in whether they used tenders, market prices or other arrangements. Basavraj would need to distinguish emergency physical-supply objectives from any ambition to profit from crude-price movements.

Will renewable energy make petroleum reserves unnecessary?

The officials acknowledged the long-term shift toward cleaner energy but argued that petroleum demand was likely to remain material for some time. That forecast raised a resource-allocation question: should India invest in additional storage while also investing in energy transition and diversification? The decision required a view of the time horizon over which oil imports would remain strategically significant.

The 2026 Geopolitical Oil Crisis: Availability Versus Price

The debate about the value of strategic oil reserves had become immediate by October 2026. The conflict in the Middle East disrupted shipping through the Strait of Hormuz and threatened other trade corridors, including routes through the Red Sea. Attacks on regional energy infrastructure, longer voyages and higher freight and insurance costs further complicated the movement of oil, even as producing countries attempted to restore exports. The International Energy Agency (IEA) reported that physical crude prices had risen sharply in September '26, and that disruptions to petroleum-product flows—especially diesel—were continuing. Its October '26 assessment found that global observed oil inventories comprised 509 million barrels, well below their end-February level; although these inventories were not identical to governments' emergency reserves [13, 14].

The supply picture was more nuanced than a complete absence of crude oil. Reuters reported on 22nd October that India's imports of Middle Eastern crude had returned to around 2.8 million barrels per day in September, approximately 54% of its total crude imports, as tanker movements through the Strait of Hormuz recovered. Russian purchases nevertheless accounted for about 1.8 million barrels per day, showing the continued importance of alternative suppliers. A cargo could again be available for purchase, but its delivery date, freight cost and effective landed price remained uncertain. The recovery in physical volumes therefore did not mean the security threat had disappeared [15].

The price impact was already visible in India's import bill. Petroleum Planning & Analysis Cell (PPAC) data put the Indian crude basket at an average US$90.19 per barrel in August 2026 and US$116.09 in September, compared with US$69.61 in September 2025. India imported 21.7 million metric tons of crude in September 2026, compared to 20.3 million tons a year earlier; the import bill increased to approximately US$15.4 billion from US$10.5 billion, indicating thereby about 7% increase in quantity that coincided with about a 47% increase in expenditure. Notably, the Indian basket was a crude-price benchmark, not the full delivered cost of every import contract or the domestic retail price of petrol or diesel [16, 17].

The international response highlighted the role—and limitations—of strategic stocks. On 11th March 2026, IEA member governments agreed to make 400 million barrels available under a coordinated emergency action. By 7th October, IEA reported that approximately 325 million barrels had been released and that members supported accelerating the balance, with priority given to diesel wherever feasible. India was an IEA Association country, not a full member subject to its mandatory 90-day stockholding obligation. Moreover, crude stored in Indian rock caverns could contribute to refinery supply, but it would not automatically resolve shortages of finished diesel if refinery operations, transport logistics or the mix of crude grades were constrained [14, 18].

These developments put two different risks in tension; a severe physical shortage might justify releasing accessible reserve oil to maintain essential refinery operations. Yet if deliveries could still be arranged at high prices, using scarce sovereign stocks to soften an import-price shock would sacrifice part of the buffer that could

be needed if a shipping route closed again. Conversely, rebuilding inventories while prices were elevated would protect future availability but increase the immediate financing burden; arrangements under which, overseas producers stored crude in Indian caverns offered another form of capacity utilization. But commercial access rights, the location and quality of the crude, and Government of India's emergency priority had to be examined before such barrels could be treated as equivalent to fully owned sovereign stock.

By October 2026, the pricing decision was therefore inseparable from the availability decision. Policymakers could preserve the reserve, and allow commercial imports to absorb higher prices; release a defined quantity of emergency crude under clear triggers; or combine limited releases with short-term procurement diversification, targeted commercial storage arrangements and future replenishment rules. However, each choice carried trade-offs for the import bill, refinery continuity, fiscal exposure and the next disruption. It was not enough to ask whether India possessed storage caverns. The more difficult question was how much deliverable oil it could control, at what cost, and under which conditions it should be deployed.

Basavraj's Team -LBOC Project Objective

After meeting the ISPRL team and seeing the scale of the facility, Basavraj was impressed by its engineering and by the difficulty of the process of acquiring land, building caverns and connecting them to terminals and refineries. Nevertheless, he questioned the financial commitment involved in constructing large underground stores and filling them with expensive crude oil. The security value of a reserve was evident; assessing its economic value was more complex.

He recalled India's 1991 balance-of-payments crisis, the consequences of the Gulf War and the country's need to mobilize foreign exchange, including by pledging gold, to maintain essential imports. **The Professor teaching in the same business school in 2nd semester in 2026 MBA, asked students** to reflect on the relationship between crude-price shocks, import costs, the current account and gross domestic product (GDP). Basavraj also reviewed a 2019 Reserve Bank of India (RBI) note estimating under that study's assumptions, that a US$10-per-barrel increase in crude prices was associated with roughly US$12.5 billion of additional trade-deficit pressure.

He decided that the MBA class had to consider two interrelated questions: what India might lose in a supply disruption without sufficient accessible emergency stocks; and whether the cost of storing oil was justified when placed against the potentially much larger macroeconomic disruption. He advised the class to plan and define the assumptions explicitly, evaluate oil-import exposure under different price scenarios, and consider approaches including historical price simulation. The accounting identity GDP = C + I + G + (X − M) offered a way to discuss external-sector exposure, but Basavraj knew that an import-price increase did not produce an identical direct change in GDP without further assumptions about consumption, substitution, taxes, exchange rates and the broader economy.

By October 2026, the problem had become more complicated. India still imported most of its crude; meanwhile, the IEA had undertaken a large coordinated release during the year, and India was pursuing both expanded domestic capacity and international storage partnerships. For Basavraj, a complete recommendation would have to examine the size of the reserve, its financing, where oil could be held, who controlled its release, and how quickly it could reach refineries in a real emergency. The case therefore retained its central question:

What is the value of holding a sovereign reserve when the most damaging energy-security event included uncertain in timing, duration and scale? The 2026 geopolitical crisis supplied a real-time test of that judgment: was an emergency reserve primarily insurance against a failure of physical supply, or could it also legitimately be deployed to alleviate a price shock while supply remained obtainable? The answer depended on whether the release left sufficient crude—and the right grades in the right locations—for a deeper interruption.

EXHIBITS Updated till October 2026

The exhibit numbering follows the original Basavraj team findings. Historical analytical examples remain clearly marked; contemporary data replace outdated industry facts. Sources listed at the end are part of the organizer-supplied evidence base.

Exhibit I. Emergency oil-stockholding models and IEA membership requirements

Model	Stock owner / operational principle	Strategic consideration
Government stock	Public entity owns physical crude or products	Budget requirement; direct public control
Agency stock	Designated agency maintains strategic stocks	Separate procurement and storage governance
Industry stock	Industry operators maintain stocks under national rules	Private operating capacity; audit and enforceability
Contractual / overseas stocks	Enforceable stock access, potentially across borders	Timing of delivery and sovereign enforceability

Source: IEA, Membership; IEA, Oil Security and Emergency Response (accessed October 2026).

Note: IEA full-member requirement: 90 days of the previous year's net imports. India is an Association country; this is not a domestic-consumption denominator.

Exhibit II. IEA coordinated emergency oil-stock actions, 1991–2026

Year	Trigger	Announced / reported action
1991	Gulf War	IEA coordinated release / response
2005	Hurricanes Katrina and Rita	Coordinated emergency response
2011	Libyan supply disruptions	Coordinated emergency response
2022 (first)	Russia–Ukraine energy crisis	Collective action
2022 (second)	Continuing energy crisis	Second collective action in 2022
11 March 2026	Middle East conflict / supply disruption	400 million barrels announced for availability

Source: IEA, Update on collective action decision of 11 March 2026; IEA, Oil Security and Emergency Response.

Supplementary Exhibit S1. Historical illustrative storage economics (original case estimates)

Storage type	Historical construction estimate (US$/bbl)	2026 interpretation
Underground rock cavern	15–31	Historic illustrative range; update through project bid documents
Underground salt cavern	8–12	Historic favorable-disposal case; brine management sensitive
Above-ground steel tanks	29–33	Historic illustrative range; land and steel cost sensitive

Source: Rajib and Basu (2022), original case based on then-available sector studies.

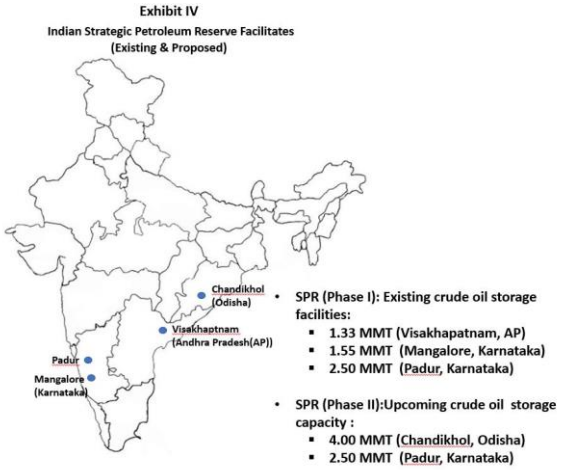
Exhibit III. India's SPR program — installed and approved facilities, October 2026

Facility	Capacity (MMT)	Position as of case date
Visakhapatnam, Andhra Pradesh	1.33	Operational; Phase I
Mangalore, Karnataka	1.50	Operational; Phase I
Padur, Karnataka	2.50	Operational; Phase I
Phase I TOTAL	5.33	Installed strategic cavern capacity
Padur II, Karnataka	2.50	Approved; LOA awarded 18 Aug 2025; target Aug 2030
Chandikhol, Odisha	4.00	Approved; state MoU Apr 2025; land process outstanding in FY25 report
Phase II TOTAL	6.50	Planned / under development, not operational

Source: PIB, Government steps to Strengthen Strategic Petroleum Reserves (20 Mar 2025); ISPRL, Annual Report FY 2024–25.

Note: The potential total of 11.83 MMT is installed plus approved capacity, not capacity available to draw in 2026.

Exhibit IV. Reserve locations (original case schematic)



Source: illustration based on ISPRL annual reports. This is a historical schematic: Phase II sites shown are approved/planned, not operational in October 2026.

Exhibit V. Construction photographs from the original ISPRL case



Source: credited to the ISPRL photo gallery. These are archival photographs, not images of Phase II work in 2026.

Exhibit VI. India petroleum exposure and reserve coverage — dated measures

Indicator	Value	Reporting period / interpretation
Crude import dependence	88.7%	FY 2025–26, PPAC-derived provisional
Crude imports	246.6 MMT	FY 2025–26, PPAC-derived provisional data vintage
Domestic crude production	27.96 MMT	FY 2025–26, stated in Parliament Aug 2026
Average Indian crude basket	US$70.99/bbl	FY 2025–26; PPAC annual average
National oil-stock capacity	~74 days	Government statement, March 2026; wider crude and product system
Actual national oil-stock cover	~60 days	Government statement, March 2026; snapshot, NOT ISPRL alone
ISPRL caverns: crude stored	~3.52 MMT	31 Mar 2025, company annual report; earlier dated stock, NOT Oct 2026

Source: PPAC FY 2025–26 Ready Reckoner / updated figures; Parliament reply Aug 2026; PIB statement March 2026; ISPRL Annual Report FY 2024–25.

Note: Provisional import figures can vary by subsequent data revision and reference period. Capacity, stock and cover days have different scopes and cannot be added or compared as like-for-like figures.

Supplementary Exhibit S3. Geopolitical oil shock — availability and pricing indicators, August–October 2026

Table S3.1 | Selected reported observations. Price levels are US dollars per barrel; Indian import quantities are million metric tons. Figures are dated, not projections.

Indicator and period	Reported value	Source and interpretation
Indian crude basket — August 2026	US$90.19/bbl	PPAC monthly average; crude benchmark, not landed cargo price [16]
Indian crude basket — September 2026	US$116.09/bbl	PPAC monthly average; Sept 2025: US$69.61/bbl [16, 17]
India crude imports — September 2026	21.7 MMT	20.3 MMT in September 2025; approximately +7% [17]
India crude import bill — September 2026	US$15.4 billion	US$10.5 billion one year earlier; approximately +47% [17]
Indian imports from Middle East — September 2026	~2.8 million bbl/day (~54%)	Reuters ship-tracking data, published 22 October; not government stock [15]
Indian imports from Russia — September 2026	~1.8 million bbl/day	Reuters ship-tracking data; supplier mix changes with pricing and flows [15]
IEA emergency oil release — status 7 October 2026	~325 million barrels released	From March action initially involving 400 million barrels; member-country releases, not Indian stocks [18]
Global observed oil inventories — end-September 2026	509 million barrels below end-February	IEA worldwide observed inventories; NOT India-specific strategic reserves [13]
India dedicated SPR installed cavern capacity	5.33 MMT	Installed capacity; NOT necessarily the quantity actually stored [05]

Source: Petroleum Planning & Analysis Cell (PPAC); Business Standard reporting of PPAC data, 21 October 2026; Reuters ship-tracking report, 22 October 2026; International Energy Agency (IEA), Oil Market Report, 14 October 2026, and statement of 7 October 2026. Full source references [13]–[18] appear in the updated register.

Reading note: The Indian basket is a quoted monthly crude benchmark, imports measure physical flows or customs quantities, and IEA emergency releases refer to member-country stocks. None should be equated with oil physically present in ISPRL caverns. July 2021 approval of Phase II adds no currently commissioned capacity.

Exhibit VII. India's 1991 balance-of-payments crisis — historical context retained

Event / measure	Historical evidence
Iraq–Kuwait conflict, 1990	Global crude-price and remittance pressures added to India's external vulnerability.
Import-cover constraint, 1991	Foreign-exchange reserves had fallen to cover only roughly two weeks of imports.

Gold pledge in July 1991	Reserve Bank of India pledged about 47 tonn of gold to raise around US$405 million in foreign-exchange liquidity.
Relevance to 2026	Illustrates the strategic cost of import vulnerability; the original crisis data are historical and are not a forecast.

Source: Reserve Bank of India, The Balance of Payments Crisis of 1991 (2018), as cited in Rajib and Basu (2022), original Exhibit VII.

Note: The 1991 figures are retained from the original case; no attempt is made to present them as 2026 data.

Supplementary Exhibit S2. India–ADNOC storage arrangements: established facts and opportunities

Date	Development	What the evidence establishes
10 Feb 2018	Mangalore storage agreement	ADNOC storage arrangement established; commercial and strategic access provisions
Sep 2024	ISPRL–ADNOC MoU	Dialogue on extending Mangalore and potential Visakhapatnam storage
15 May 2026	Strategic collaboration announcement	Potential storage up to 30 million barrels in India; possible Visakhapatnam / Chandikhol / Fujairah cooperation

Source: ISPRL, Annual Report FY 2024–25; PIB, List of Outcomes: Prime Minister's Visit to the UAE, 15 May 2026; ADNOC press release, 15 May 2026.

Note: The 2026 announcement describes opportunities rather than 30 million barrels of physical stocks already in place.

Exhibit VIII. Oil-price shock sensitivity: original RBI rule and FY 2025–26 import exposure

Approach	Input	Indicative result / limitation
RBI Mint Street Memo (2019)	US$10/bbl oil-price rise	US$12.5bn additional deficit estimate under 2019 model assumptions
FY26 import-bill arithmetic	246.6 MMT × 7.33 barrels/ton × US$10	~US$18.08bn annual gross import bill sensitivity, holding volume and mix fixed
Difference between estimates	2019 macro study vs 2026 static volume exercise	NOT competing measurements; exchange rates, demand response and export offsets differ

Source: RBI, Ghosh & Tomar (2019), The Impact of Crude Price Shock...; PPAC FY 2025–26 import figures.

Note: The barrel conversion factor of 7.33 is an approximation; crude-grade mix affects actual barrels per ton. Static arithmetic is not a GDP forecast.

Exhibit IX. Crude-to-petrol arithmetic for FY 2025–26

Input / step	FY 2025–26 illustrative value	Basis
Average Indian crude basket	US$70.99 per barrel	PPAC annual Indian-basket average
Average ₹/US$ rate	₹88.31 per US$	FY 2025–26 RBI average, rounded
Gross crude equivalent per liter	₹39.43	70.99 × 88.31 ÷ 159 liter
Delhi petrol price, annual mean	₹94.77 per liter	PPAC historical Delhi average

Source: PPAC Ready Reckoner FY 2025–26; RBI exchange-rate annual average (as reproduced with the PPAC series).

Note: Gross crude equivalent is NOT retail petrol price; taxes, refining, blending, dealer margin and product yields differ. The original 2022 exhibit used a historical ₹70/US$ assumption, not a current exchange rate.

Exhibit X. Selected crude-price reference points for the case

Measure	Price (US$/bbl)	Period / consistency
Brent crude (original case)	61.63	Jan 2020 monthly historical series; unchanged archival example
India crude basket	70.99	FY 2025–26 average; Indian basket, not Brent
India crude basket	98.01	March 2026; reported monthly average
India crude basket	116.09	September 2026 monthly average

Source: US EIA, Brent monthly historical series; PPAC Indian crude basket data published by Oct 2026.

Note: Do not combine Brent historical values and Indian-basket monthly observations into a statistically uniform time series without a benchmark adjustment.

Exhibit XI-A. Historical US Treasury yields — original analytical input retained

Year	1-year YTM	10-year YTM	20-year YTM	30-year YTM
2010	0.32%	3.21%	4.00%	4.25%
2011	0.18%	2.78%	3.46%	3.91%
2012	0.17%	1.80%	2.54%	2.92%
2013	0.15%	2.35%	2.68%	3.08%
2014	0.18%	2.14%	3.53%	3.78%
2015	0.32%	2.14%	2.55%	2.84%
2016	0.61%	1.84%	2.22%	2.59%
2017	1.20%	2.33%	2.65%	2.89%
2018	2.33%	2.91%	3.02%	3.11%
2019	2.05%	2.14%	2.40%	2.58%

Source: Original AIMA–ICRC 2022 case, Exhibit XI (US Treasury annual yield data, 2010–2019).

Note: Archival figures are intentionally retained, not updated 2026 yields. They must not be represented as a current risk-free curve. Participants may instead analyze the updated scenario in Exhibit XI-B.

Exhibit XI-B. Illustrative FY 2025–26 import-price sensitivity (not an oil-price forecast)

Price change vs baseline (US$/bbl)	Fixed annual import quantity (MMT)	Estimated gross bill change (US$bn)
-10	246.6	-18.08
-5	246.6	-9.04
0	246.6	0
+5	246.6	+9.04
+10	246.6	+18.08
+20	246.6	+36.15

Source: Organizer calculation using PPAC import volume (246.6 MMT provisional) and 7.33 barrels per ton.

Note: This exhibit replaces the dated 2010–19 US Treasury yield grid as the immediately relevant 2026 sensitivity. The original historical Treasury series may still be consulted in the 2022 source case for the time-value / Monte Carlo discussion.

References and Data-Verification Register

Original case reference: Rajib, P., and Basu, S. (2022), "ISPRL: Managing India's Sovereign Crude Oil Price Risk," AIMA–India Case Research Centre, AIMA/ICRC/2022/051. Historic conversations, engineering descriptions and original historical cost figures derive from this source; subsequent facts below are separately dated.

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